

Report No.	SD-LAB-2025-0847	Sample Received	20 Sep 2025
Batch ID	PRO100-SEP25-B12	Analysis Date	22 Sep 2025
Product	Maize Silage Bale — Pro 100 (100kg)	Report Date	23 Sep 2025
Harvest Date	18 Sep 2025	Lab Reference	NABL/2025/4471

OVERALL RESULT

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MEETS QUALITY

STANDARDS

Fermentation: GOOD | Stability: STABLE

| Safety: PASS

1. Proximate Analysis (As-Fed Basis)

Parameter	Unit	Result	Typical Range	Status
Dry Matter (DM)	% as-fed	34.8	30.0 – 38.0	PASS
Moisture	% as-fed	65.2	62.0 – 70.0	PASS
Crude Protein (CP)	% DM	8.9	7.0 – 10.5	PASS
Neutral Detergent Fibre (NDF)	% DM	44.2	38.0 – 52.0	PASS
Acid Detergent Fibre (ADF)	% DM	24.8	20.0 – 30.0	PASS
Starch	% DM	33.1	28.0 – 38.0	PASS
Crude Fat (EE)	% DM	3.4	2.5 – 4.5	PASS
Crude Ash	% DM	3.8	3.0 – 5.5	PASS
Metabolisable Energy (ME)	MJ/kg DM	11.4	10.5 – 12.5	PASS
Net Energy Lactation (NEL)	MJ/kg DM	6.8	6.2 – 7.5	PASS

2. Fermentation Quality Analysis

Parameter	Unit	Result	Acceptable Range	Status
pH	—	3.9	3.5 – 4.2	PASS
Lactic Acid	% DM	5.8	4.0 – 8.0	PASS
Acetic Acid	% DM	1.9	0.5 – 3.0	PASS
Propionic Acid	% DM	0.12	< 0.5	PASS
Butyric Acid	% DM	< 0.01	< 0.10	PASS
Ammonia-N (NH3-N)	% of Total N	5.2	< 10.0	PASS
Lactic:Acetic Ratio	—	3.1	> 2.0	PASS
Water Activity (Aw)	—	0.97	0.94 – 0.99	PASS

Fermentation Profile: Predominantly lactic acid fermentation with minimal acetic acid. Butyric acid below detection limit — no clostridial activity. Low ammonia-N confirms minimal protein degradation. Overall fermentation quality: EXCELLENT.

3. Physical Characteristics & Hygiene Assessment

Parameter	Observation / Result	Standard	Status
Colour	Yellow-green, uniform	Yellow to olive green	PASS
Odour	Pleasant, mild acidic	Fruity/acidic, no foul odour	PASS
Texture	Chop length 8–12 mm	6–15 mm theoretical chop	PASS
Visible Mould	None detected	Absent	PASS
Foreign Material	None detected	Absent	PASS
Wrapping Integrity	6 layers, no punctures	Min 4 layers	PASS
Aerobic Stability	> 120 hours	> 72 hours	PASS
Total Plate Count	< 10,000 cfu/g	< 50,000 cfu/g	PASS
Yeast & Mould Count	< 500 cfu/g	< 5,000 cfu/g	PASS

4. Feeding Recommendations

Animal Category	Recommended Inclusion	Expected Benefit
Milking Cows (HF/Jersey)	8 – 12 kg/head/day	Supports 18–22 L milk/day; improves milk fat %
Dry Cows	4 – 6 kg/head/day	Body condition maintenance; rumen conditioning
Heifers (12–24 months)	3 – 5 kg/head/day	Frame growth; rumen development
Calves (> 4 months)	1 – 2 kg/head/day	Introduction to forage; rumen training
Buffalo (Milking)	10 – 14 kg/head/day	Supports 8–12 L milk/day; improves digestibility

Note: Inclusion rates are indicative. Adjust based on animal body weight, milk yield targets, and total ration composition. Introduce silage gradually over 7–10 days. Ensure fresh water is always available.

5. Storage & Handling Guidelines

Storage Temperature	Store below 35°C. Avoid direct sunlight on wrapping.
Shelf Life (Sealed)	12 months from harvest date under proper storage conditions.
After Opening	Use within 2–3 days. Reseal cut face tightly after each feeding.
Wrapping Damage	Inspect for punctures before use. Patch immediately with silage repair tape.
Feeding Area	Remove uneaten silage within 4 hours in hot weather (> 30°C).
Signs of Spoilage	Discard if black/grey colour, butyric odour, or extensive mould growth.

Analysed By	Verified By	Laboratory
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Feed Quality Assurance — SilageDirect	SilageDirect, Mehsana, Gujarat	Report valid for 12 months

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